

CYBER INTELLIGENCE + ETHICS

Presented By: **WICYSUMD**



INTRODUCTION



Goals of the Workshop:

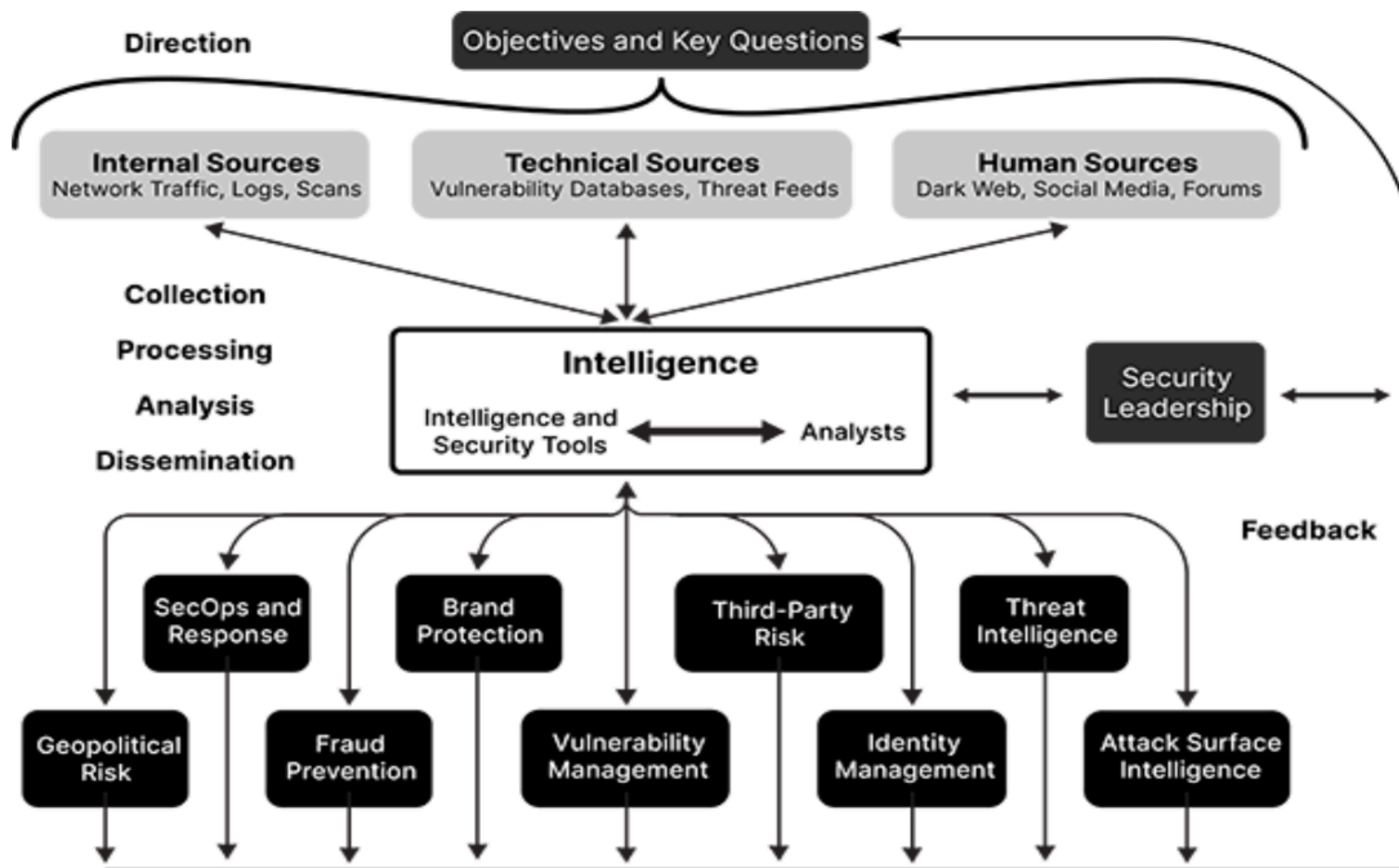
- Understand ethical considerations in cyber intelligence
- Learn ethical frameworks for cybersecurity decision-making
- Apply ethical principles to real-world scenarios



WHAT IS CYBER INTELLIGENCE?

Information that is actionable for a specific audience that allows you to make decisions in preventing or mitigating cyber threats

INTELLIGENCE LIFE CYCLE



Source: <https://www.agari.com/wp-content/uploads/2021/05/cyber-threat-intel-blog-lifecycle.png>

TYPES OF CYBER INTELLIGENCE

- | | |
|------------------|-------------------|
| 1. Vulnerability | 5. Fraud |
| 2. Third Party | 6. Attack Surface |
| 3. Brand | 7. Counter-Intel |
| 4. Geopolitical | |

OPERATIONAL VS STRATEGIC INTEL

1. **Operational/Technical** --> knowledge about ongoing cyberattacks, events, and campaigns.
2. **Strategic** --> provides a broad overview of an organization's present and future threat landscape

INTELLIGENCE THREATS - KNOWING YOUR ENEMY

1. Organized Crime Group
2. Gated Communities
3. State-sponsored hacking groups
4. Hacktivists
5. Cyber criminals

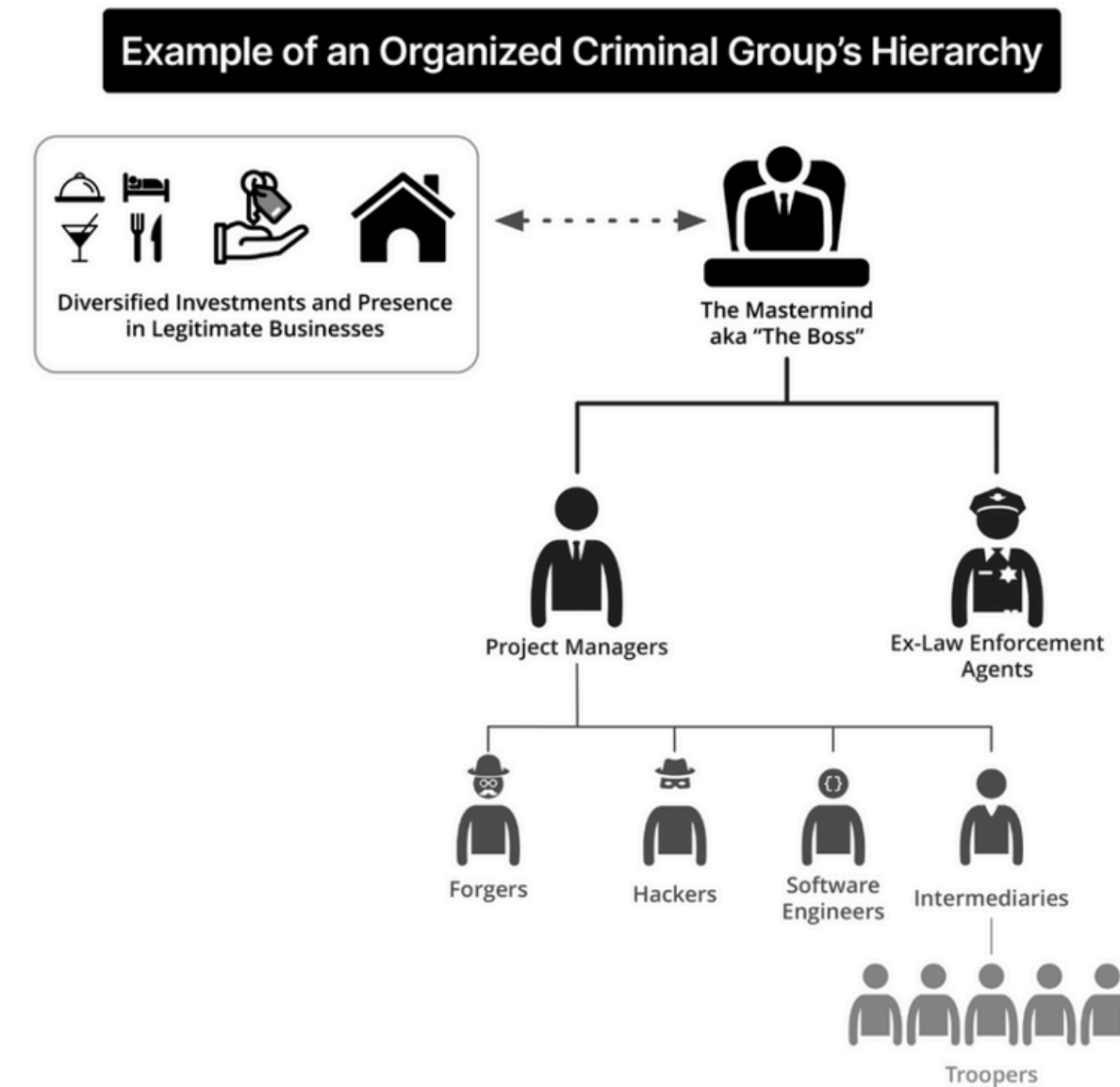


Figure 7-2: A typical organizational chart for a cybercrime syndicate. (Source: Recorded Future)

INTELLIGENCE FACTORS - RISK ANALYSIS

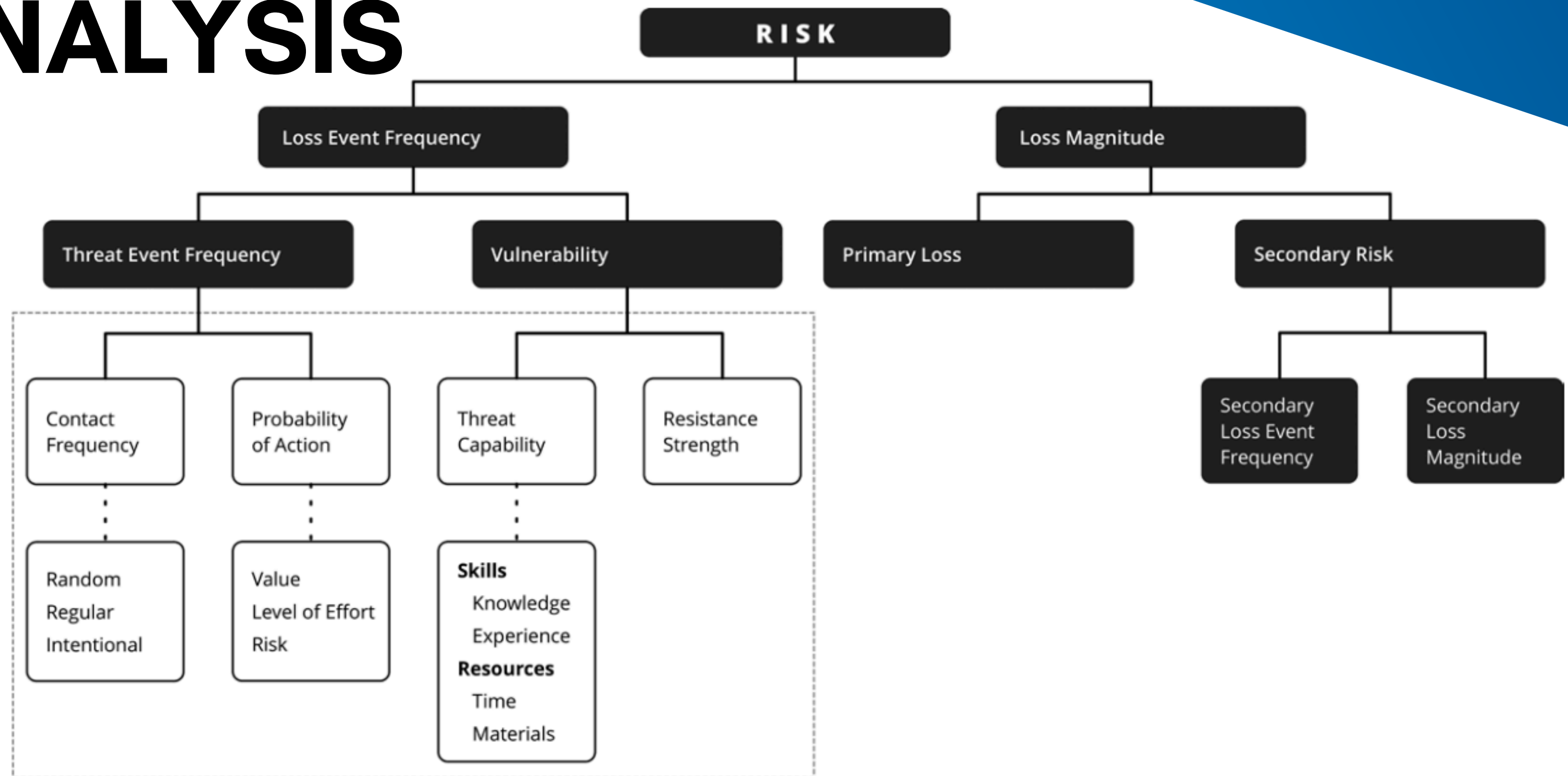
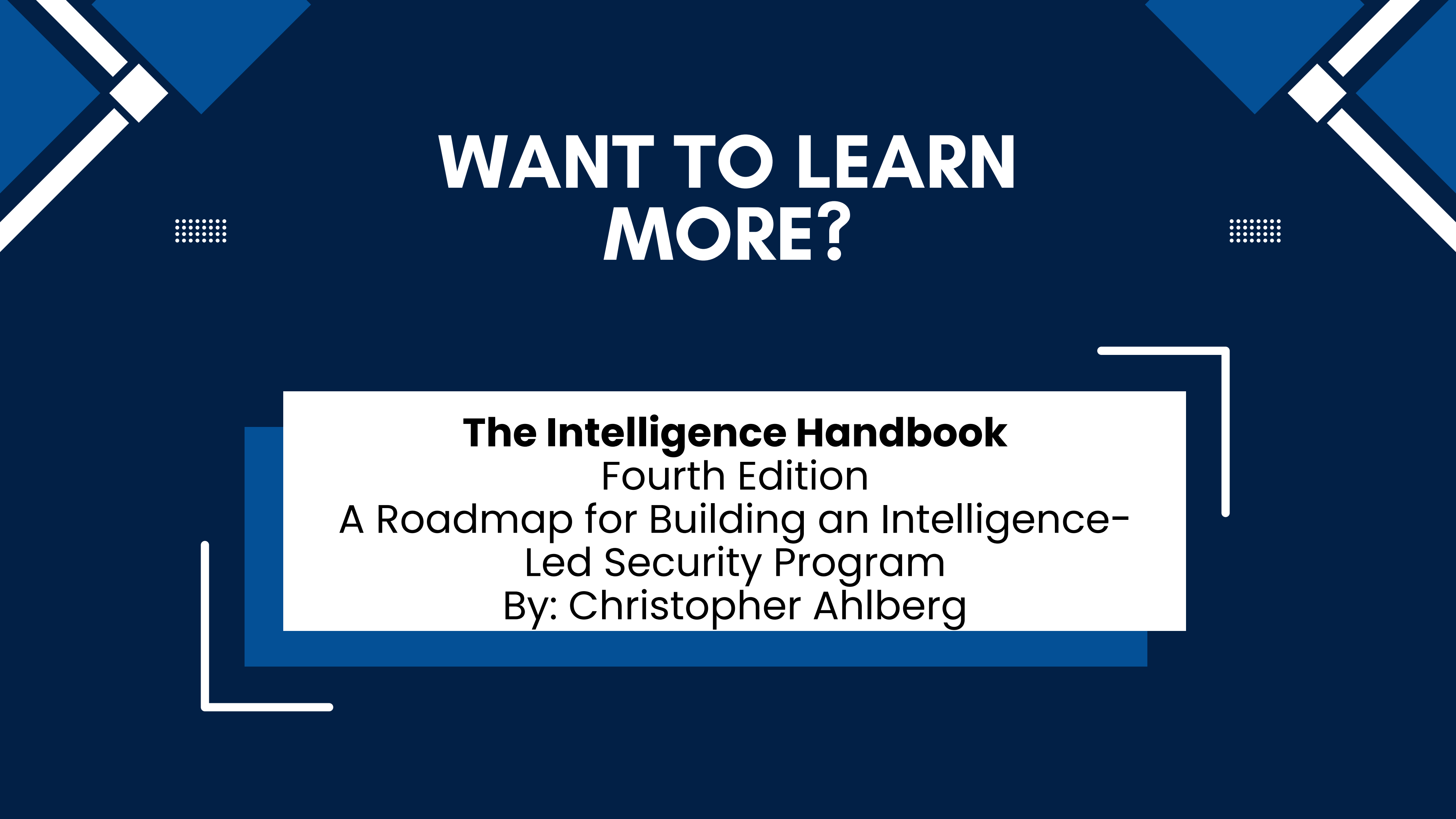


Figure 8-1: The FAIR Framework, with elements informed by intelligence highlighted. (Source: The FAIR Institute)



WANT TO LEARN MORE?

The Intelligence Handbook

Fourth Edition

A Roadmap for Building an Intelligence-
Led Security Program

By: Christopher Ahlberg



ETHICAL FOUNDATIONS IN CYBERSECURITY

What is Ethics?

Why Ethics Matter in Cyber Intelligence



What is **Ethics**?



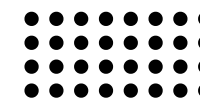
A set of **moral principles or values** that guide how people should behave, distinguishing right from wrong, and often used to define what is considered good or bad conduct.



Why **Ethics** Matter in **Cyber Intelligence**

Ensures responsible data handling, protects privacy, maintains trust, and prevents potential harm, ultimately contributing to a more secure and ethical digital environment.

ETHICAL HACKING VS. **MALICIOUS** HACKING



- Hacking refers to gaining unauthorized access to systems, networks, or data, but not all hacking is unethical.
- Ethical hacking is a legal and responsible way to test security, while malicious hacking is done with harmful intent.

ETHICAL HACKING (WHITE-HAT)

Ethical hacking involves authorized security testing to identify and fix vulnerabilities.

Purpose

- Strengthen cybersecurity defenses
- Prevent cyberattacks
- Help organizations comply with regulations

Who Performs It?

- Certified Ethical Hackers (CEH)
- Penetration Testers
- Security Researchers

Example: A company hires an ethical hacker to conduct penetration testing to find weaknesses before attackers exploit them.

BLACK HAT HACKING

Malicious hacking involves unauthorized access to exploit, steal, or damage data and systems.

Purpose

- Financial gain (e.g., ransomware, credit card fraud)
- Political motives (e.g., cyber espionage)
- Personal revenge (e.g., insider threats)

Example: A hacker infiltrates a hospital's database, encrypts patient records, and demands ransom.

GREY HAT HACKING

A gray-hat hacker operates without permission but doesn't necessarily have malicious intent.

Common Scenarios:

- Finds vulnerabilities and reports them without authorization
- May demand compensation for revealing security flaws

DISCUSSION:

Should companies be forced to reward gray-hat hackers for uncovering security risks?



REAL-WORLD ETHICAL SCENARIOS (EXERCISE)

SCENARIO

A company hires you as an ethical hacker to test its cybersecurity defenses. During your testing, you discover a severe vulnerability that could allow attackers to steal sensitive customer data. However, fixing the vulnerability would require significant downtime for the company's operations, potentially resulting in financial losses.

QUESTIONS

- What steps would you take to address this situation responsibly?**
- How would you balance the ethical duty to protect customer data with the company's financial concerns?**

SCENARIO

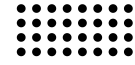
You are analyzing cyber threat intelligence and discover that an organized crime group is planning an attack on a competitor of your client. Sharing this information could help your client gain a competitive advantage but might also expose your intelligence-gathering methods.

QUESTIONS

- **Should you share this information with your client? Why or why not?**
- **How do ethical principles guide your decision in this scenario?**

POSITIONS IN CYBER INTEL & ETHICS

- **Threat Intelligence Analyst**
- Cyber Intelligence Specialist
- **Digital Forensics Analyst**
- Ethical (White Hat) Hacker
- **Cybersecurity Lawyer**
- Governance, Compliance & Risk (GRC) Manager
- **Security Awareness Trainer**



THANK YOU
FOR YOUR ATTENTION
AND PARTICIPATION

